

EE115B *Digital Circuits*

Instructor: Chenxi Xiao

Part of slides are from

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Hengzhao Yang

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Yajun Ha



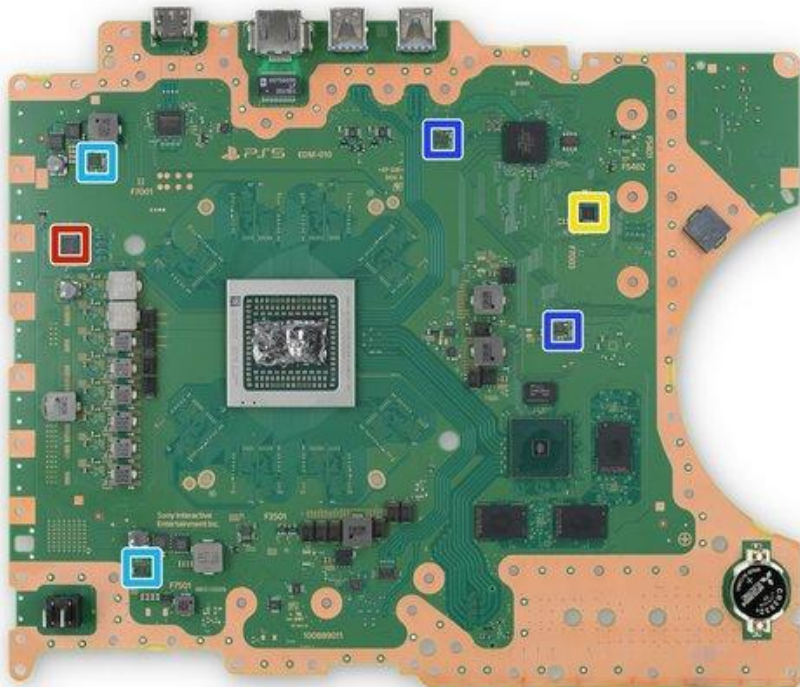
上海科技大学
ShanghaiTech University

Part 1

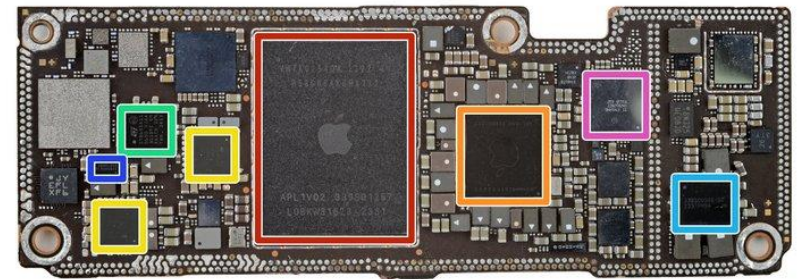
Course Information

Why digital circuits?

Digital chips are so commonly used everywhere.



<https://www.ifixit.com/Teardown/PlayStation+5+Teardown/138280>



<https://www.ifixit.com/Guide/iPhone+15+Pro+Max+Chip+ID/165320>



Why digital circuits?

For computational purposes: Processors, RAM, Flash, ...

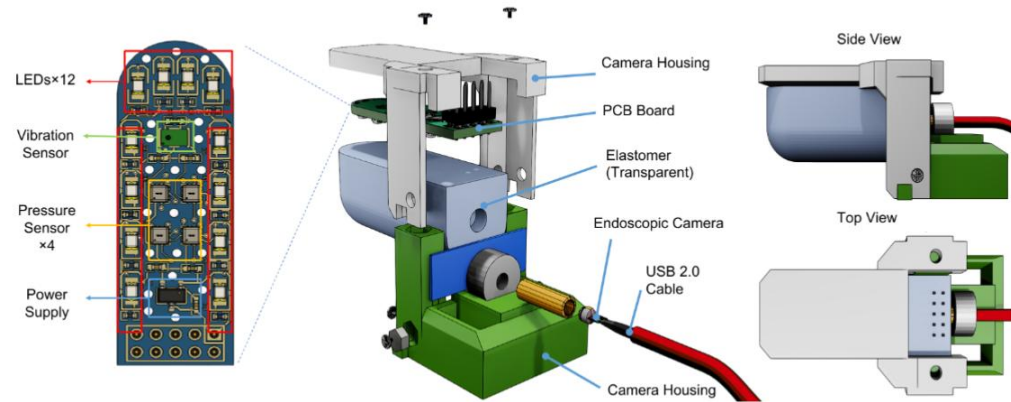


Many Systems are Hybrid

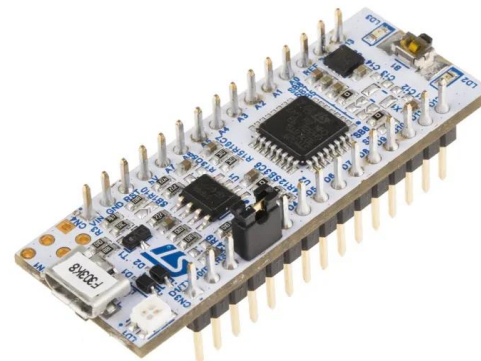
Robots



Sensors



MCU with ADC/DAC



Earphones



The position of this course

A **fundamental**, **EE** course

- EE111 Electric Circuits
- EE115A Analog Circuits
 - => EE112 Analog Integrated Circuits
- EE115B Digital Circuits
 - => EE113 Digital Integrated Circuits
 - => Embedded Systems
 - => FPGA ...
 - => Board level circuit design



Course Information

Course Instructor: Chenxi Xiao 肖晨曦

Office: SIST 1D-301A

email: xiaochx@shanghaitech.edu.cn

我的课表

打印

学年学期: 2026-2027学年1学期 切换学期

节次/周次	星期一	星期二	星期三	星期四	星期五	星期六	星期日
第一节 8:15 - 9:00	数字电路(EE115B.02) (1-15,教学中心102)			数字电路(EE115B.02) (单1-15,教学中心102)			
第二节 9:10 - 9:55	数字电路(EE115B.01) (1-15,教学中心102)			数字电路(EE115B.01) (单1-15,教学中心102)			
第三节 10:15 - 11:00							
第四节 11:10 - 11:55							

Odd Weeks

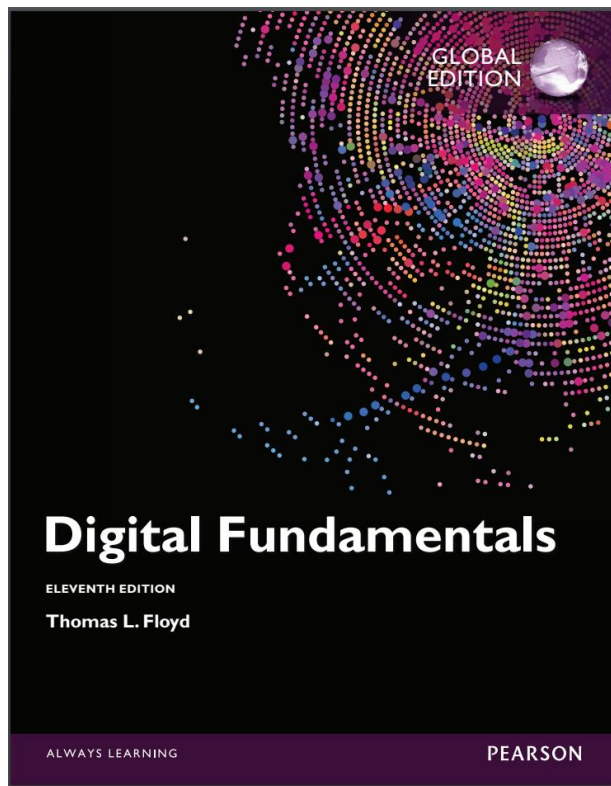
Experimental Course Instructor: Juan Li 李娟

Office: 1B-204

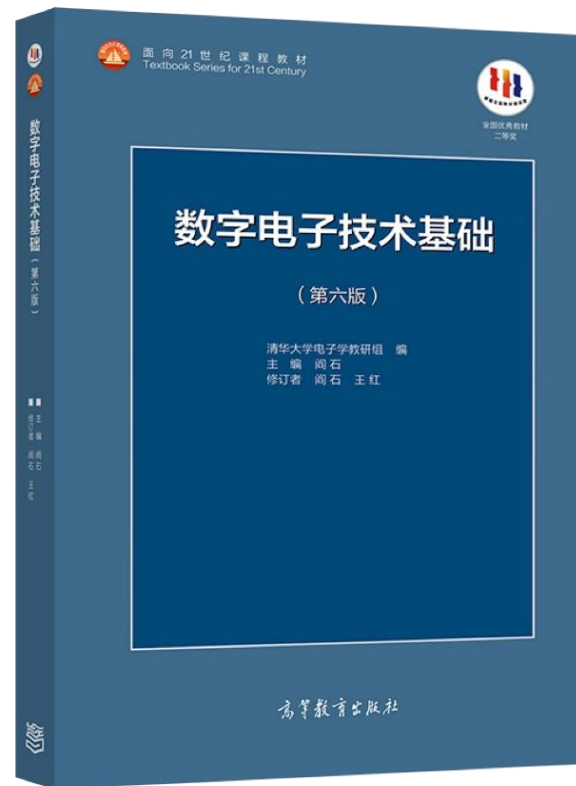
email: lijuan1@shanghaitech.edu.cn



Textbooks



Thomas L. Floyd,
Digital Fundamentals



阎石
数字电路技术基础



Syllabus

Chapter Name	Main Teaching Content (Main Knowledge Points)
1	Numbering, Coding, Logic Gates
2	Boolean Algebra and Logic Simplification
3	Logic Conversion
4	Combinational Logic Circuits: Adders, Comparators, Code Converters, Multiplexers, etc.
5	VHDL Hardware Description Language Overview
6	Sequential Logic Circuits: Latches, Flip-Flops, Counters, Registers, State Machines
7	Data Storage Circuits
8	Signal Processing Circuits

Grading Scheme:

- Attendance: 5%
- Homework: 25%
- Laboratory: 15%
- Midterm Exam: 25%
- Final Exam: 30%



Grade:

- Letter grade is based on ranking.

Integrity

- Do not copy other people's homework.
- Responsible use of tools (Including AI)

Note: except for programming/software-use homeworks, homework problems needs to be hand written.



Office Hours

Office Hour: Mon 9:55 am - 10:25 am

Mail appointment based, Immediately after class

TAs: appointment based. Send email first:

(may subject to changes by 2nd weekend)

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李建昊 lijh2024@shanghaitech.edu.cn

郭浩然 guohr2026@shanghaitech.edu.cn

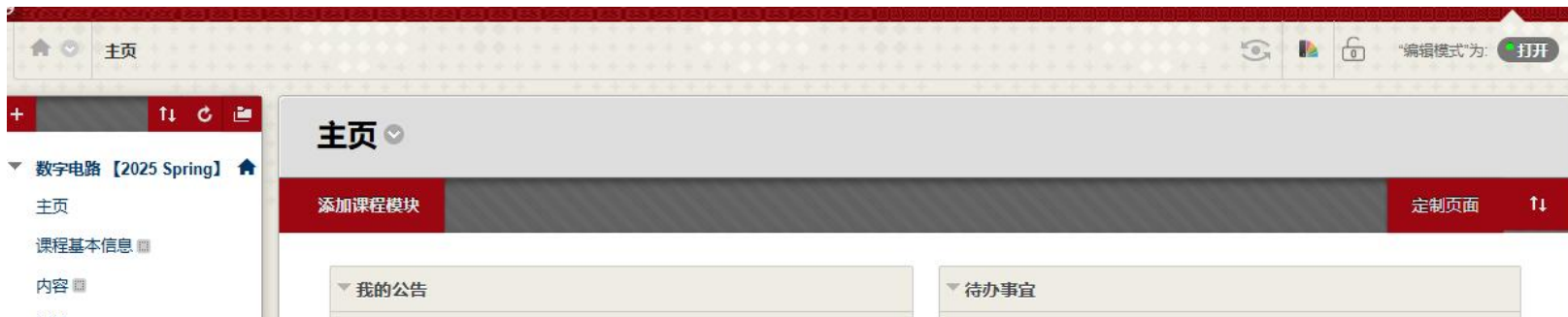
燕昱冰 yanyb2023@shanghaitech.edu.cn

黄一 huangyi2024@shanghaitech.edu.cn



Blackboard:

- Official announcements
- Publish course slides
- Homework submission



Feishu Group

- General, unofficial QA
- No important announcements
- (for grade-related questions, please inquiry via email, by Week 17)



Chapter 1: Number Systems and Coding

Digital quantity, binary, octal, hex, ...

<u>Decimal</u>	<u>Binary</u>	<u>Octal</u>	<u>Hex</u>
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

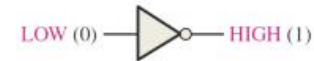
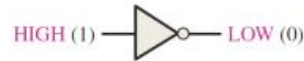


Chapter 2-3: Boolean Algebra, Logic Circuit Simplification

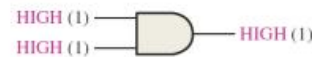
1. basic logic

2. logical expressions

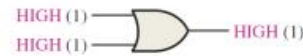
& simplification



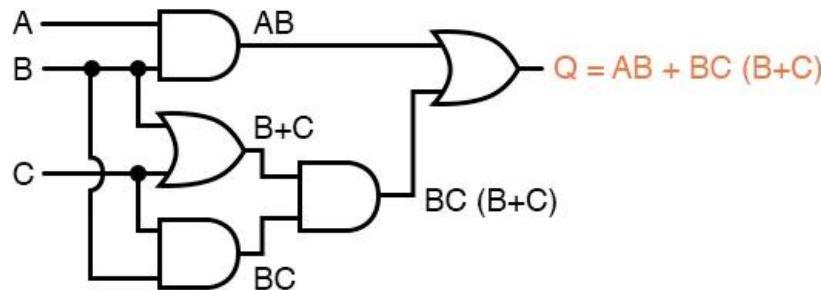
NOT gate (Inverter)



AND gate



OR gate



$$AB + BC(B + C)$$



Distributing terms

$$AB + BBC + BCC$$



Applying identity $AA = A$ to 2nd and 3rd terms

$$AB + BC + BC$$



Applying identity $A + A = A$ to 2nd and 3rd terms

$$AB + BC$$



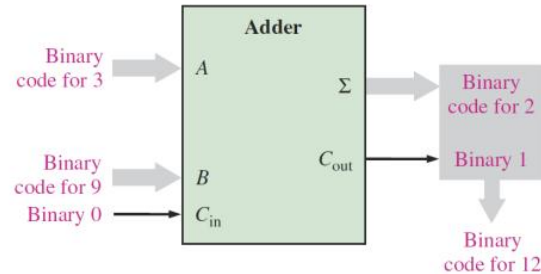
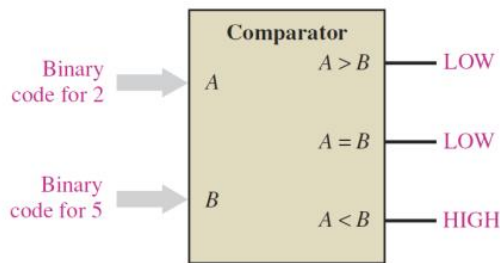
Factoring B out of terms

$$B(A + C)$$

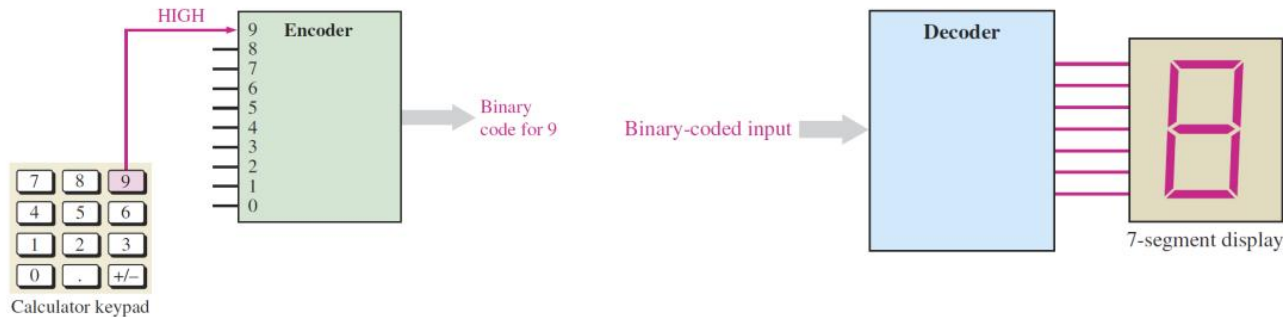


Chapter 4: Combinational Logic

- The output only dependent on the present input



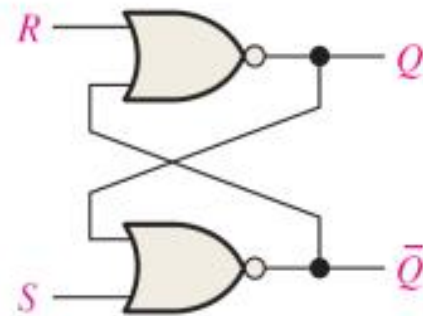
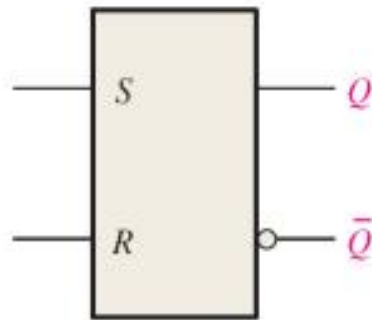
Subtraction
Multiplication
Division...



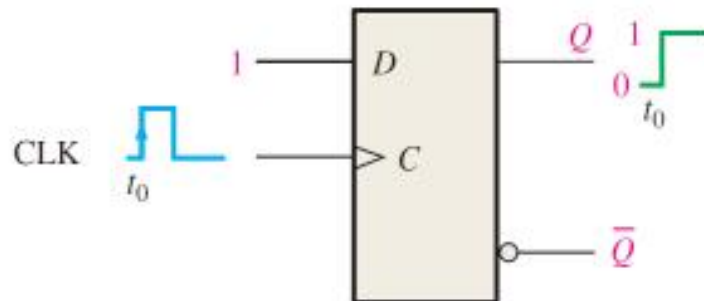
Chapter 6: Sequential Logic

- The output depends not only on the present input but also on the history of the input

Latch

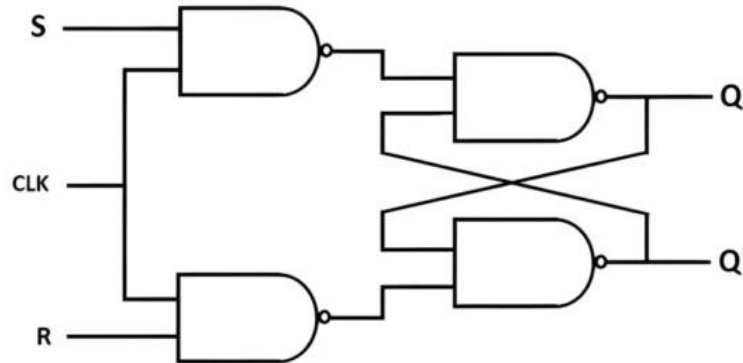


Flip-Flop

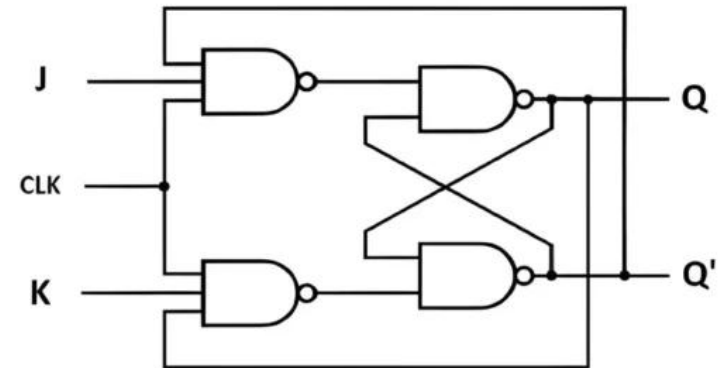


Chapter 7: Storage

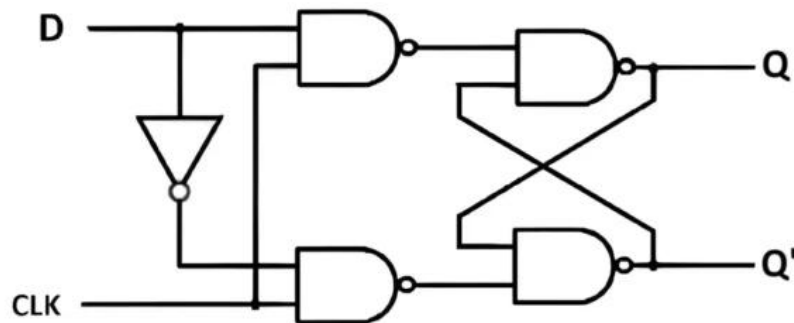
SR Flip Flop



JK Flip Flop

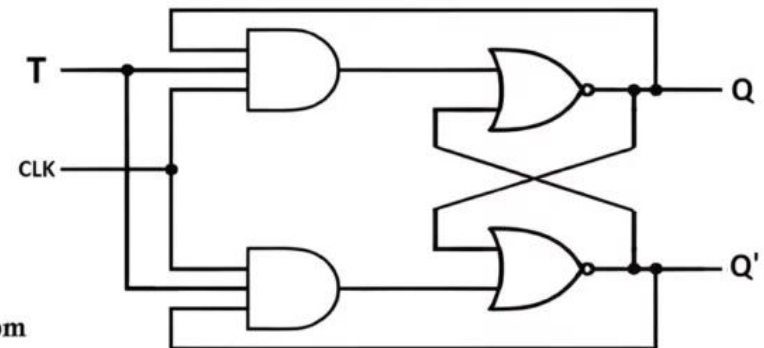


D Flip Flop



hackatronic.com

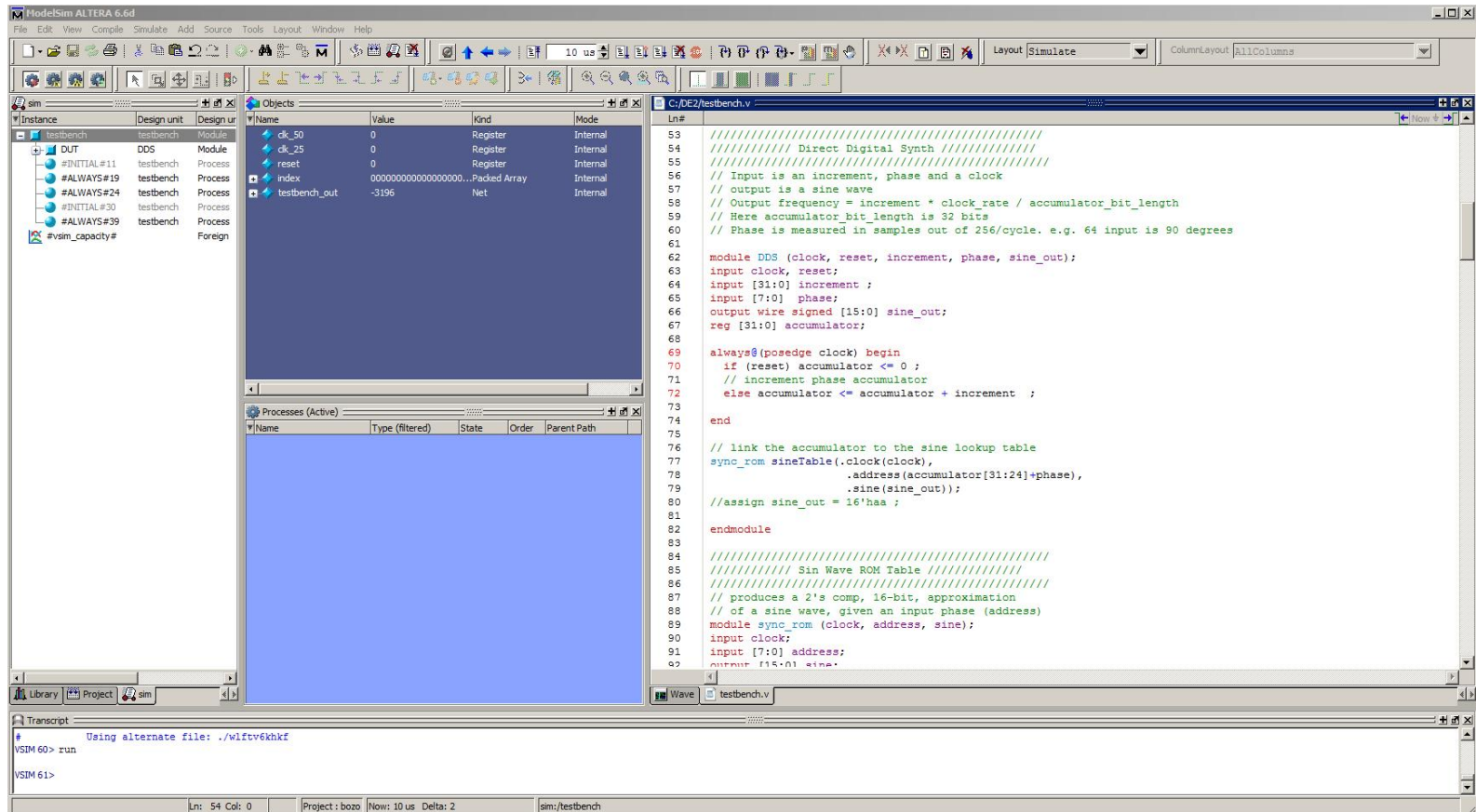
T Flip Flop



<https://www.hackatronic.com/what-is-flip-flop-circuit-truth-table-and-various-types-of-flip-flops/>



Chapter 5 Hardware Description Language



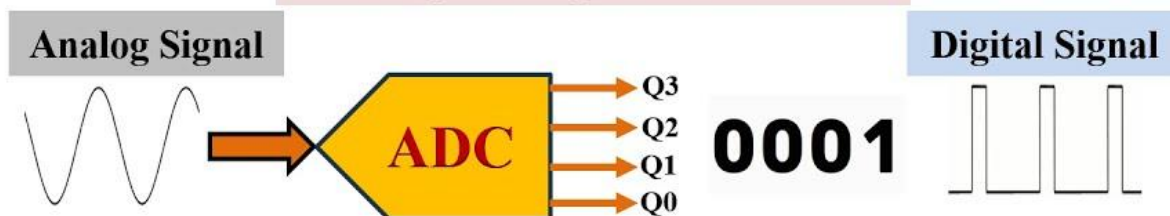
<https://people.ece.cornell.edu/land/courses/ece5760/MODELSIM/index.html>



Chapter 8: Signal Conversion

ADC & DAC

Analog To Digital Converter



Digital To Analog Converter

